

SUMMARY

Photonics engineer with 8 years of experience across leading-edge hardware systems, integrated-photonics, quantum devices, high-speed electronics, VLSI design tooling and digital signal processing. Drove system architecture, characterisation and integration of TFLN electro-optic devices, with hands-on PIC design and tape-out experience. Built end-to-end products in quantum networking and robotics industry.

SKILLS

Photonics & quantum circuitry. Test-driven PIC design for 1 to 1.5 μm wavelengths involving modulators (EOMs), phase shifters, resonators, quantum photonic circuits; precision metrology techniques, polarimetry, interferometry

Experiment design & automation. Chip and wafer-level test-station bring-up for electro-optic devices; Python instrument control and data acquisition; NI LabVIEW; ARTIQ experiment control for trapped-ion systems; tunable lasers, VNAs, spectrum analysers, power meters, AWGs, RF drivers, oscilloscopes, probe station, helium-pumped SNSPD / cryogenic and vacuum systems;

Platforms & tools. TFLN (HyperLight, Luxtelligence) and silicon (Cornerstone, AIM) PDKs, Lumerical, IPKISS, GDSfactory, KLayout, COMSOL; Python (PyTorch/DSP), MATLAB, C/C++, SystemVerilog/UVM verification pipelines

System modelling. Quantum-optical circuit modelling; cross-domain requirements (optical/RF/thermal/mechanical); optical link budgeting; latency and energy-efficiency trade-offs; entanglement swapping and time-domain interleaving

EXPERIENCE

Nu Quantum — Senior Photonics Engineer

Cambridge, 2023 – Present

- Modelled photonic circuits in TFLN PDKs and developed precision metrology for ultra-low loss ($<0.2\text{ dB}$) components, high extinction switching, polarimetry and photon-statistics to build and validate quantum photonic circuits (>0.997 fidelity entanglement distribution).
- Led system and component-level test and validation of a packaged TFLN PIC ($<4\text{ dB}$ loss, $>40\text{ dB}$ crosstalk, bounded drift) for wafer- and die-level PIC characterisation across two tapeout generations; improved test throughput/speed by $>10\times$
- Built automated measurement pipelines and instrument control in Python and ARTIQ (tunable lasers, power meters, SNSPD counters, probe station, RF drivers, AOMs) for acousto-optic pulse shaping in atomic experiments and electro-optic entanglement switching
- Performed root-cause analysis on performance deviations and fed characterisation results back into device design and roadmap via collaboration with architects, product owners and partners.
- Installed and commissioned probe station, vacuum and cryogenic systems, ran the optics lab; evaluated vendors, drove requirements, defined instrumentation budgets, hired and mentored engineers.

Photonics research student

EU (various), 2021 – 2023

- ICFO — Barcelona:** built a free-space optical monochromator and characterised entangled photon sources.
- Ghent University — Belgium:** designed and characterised a 1550 nm Si-Ph vernier resonator for refractive-index extraction and wafer statistics.
- Quandela — Paris:** LOQCathon2022: Implemented a quantum reinforcement learning algorithm for nanophotonic switch on quantum circuit simulator and processor.
- Vrije Universiteit Brussel:** Modelled and characterised dynamical response of multi-wavelength PIC laser for optical-to-THz signal conversion (10 Gb/s @ $<1e-3$ BER).

Siemens EDA — Lead Member Technical Staff, DRAM IP Verification

Noida, 2018 – 2021

- Led DDR5/DDR4 verification IP for a flagship EDA product line: JEDEC-compliant checkers, coverage, and statistical performance characterisation (throughput, timing, power) in SystemVerilog, UVM and C++ — deep test-automation and data-analysis rigour across DDR2–DDR5.

Grey Orange Robotics — Electronics Engineer

Gurgaon, 2016 – 2018

- Led EMI/EMC testing, CE certification, RF/PCB bring-up, and embedded system interfacing for Power-over-Ethernet warehouse electronics across the full product lifecycle.

EDUCATION

M.Sc. Photonics Engineering — Greatest Distinction

Belgium, 2021 – 2023

Universiteit Gent & Vrije Universiteit Brussel

B.E. (Honors) Electronics & Instrumentation Engineering

2012 – 2016

Pilani, BITS Pilani

Foundry training

Across UK, 2024 – 2025

Fabless PIC Design (AIM Photonics); PIC Bootcamp (Cornerstone Foundry)

PUBLICATIONS & PATENTS

Patent. Compressive-sensing sparse-sampling PPG measurement — US P95539.

Journal. Design of a High-Gain Two-Stage CMOS Operational Amplifier — *IJIEASR* 4(6), 2015.